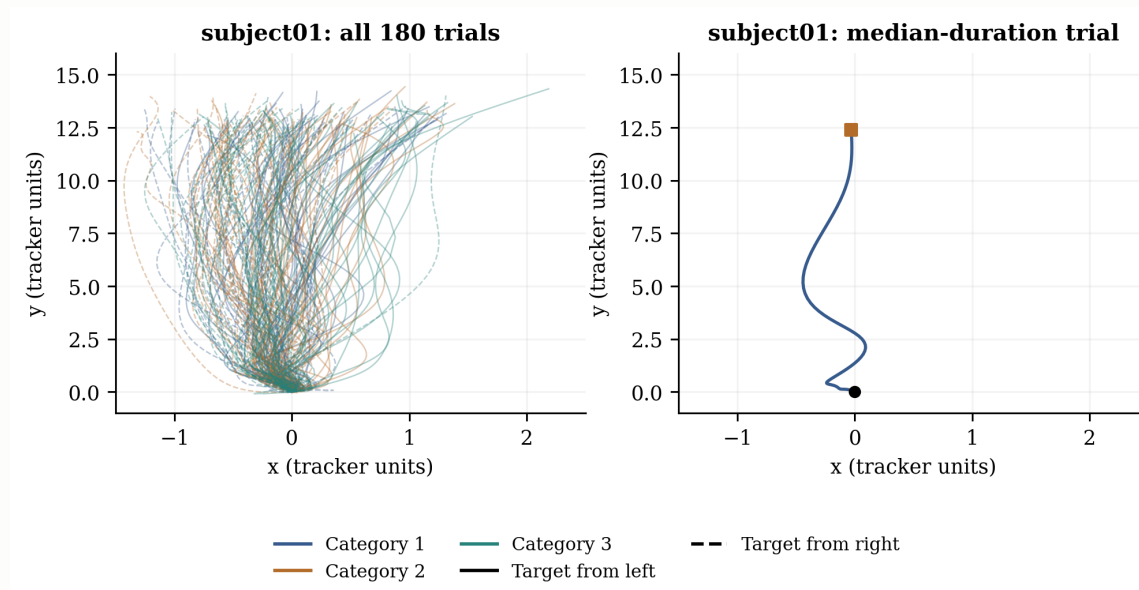


Compact Representations of Human Interception Movements

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- Supervision: Prof. Jason Friedman | Advisor: Moni Shahar



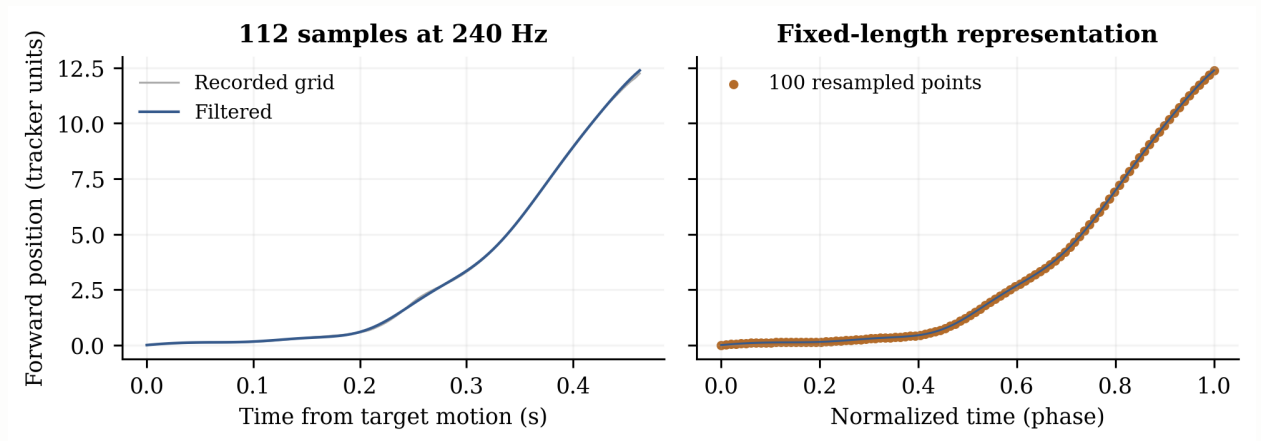
THE QUESTION

Everyone moves differently. Can we compress that?

- Task: intercept a moving target. Fast, ballistic, under one second.
- 28 participants, 4,732 trials, finger position at 240 Hz.
- Everyone solves it, but no two people move the same way.

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numbers per person?

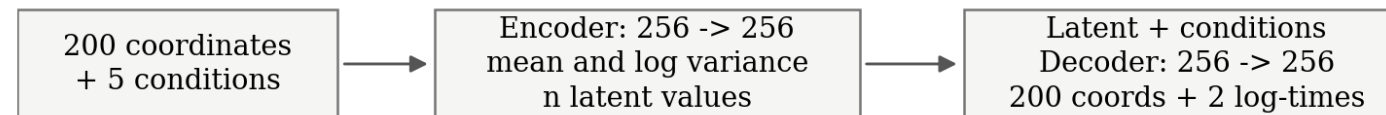


Target is a distribution of movements, not one trajectory - perception and movement are stochastic.

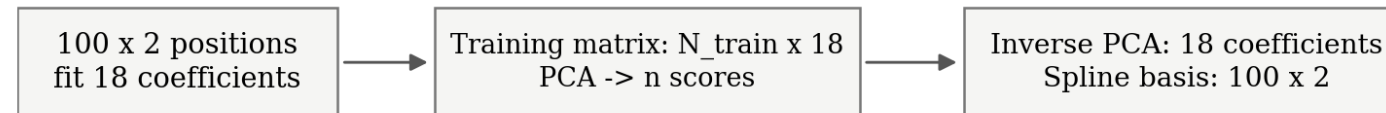
Five models, one protocol

- Every trial compressed to n numbers, then decoded back.
- Spline + PCA (linear, $\sim 10^2$ parameters) vs VAE (neural, $\sim 10^6$).
- Plus a conditional VAE, a conditional autoencoder, and a task-only floor.
- 4 folds x 17 train / 4 validation / 7 test participants.
- Each participant held out exactly once. Nothing is tested on its own training data.

Conditional variational autoencoder



Spline + PCA



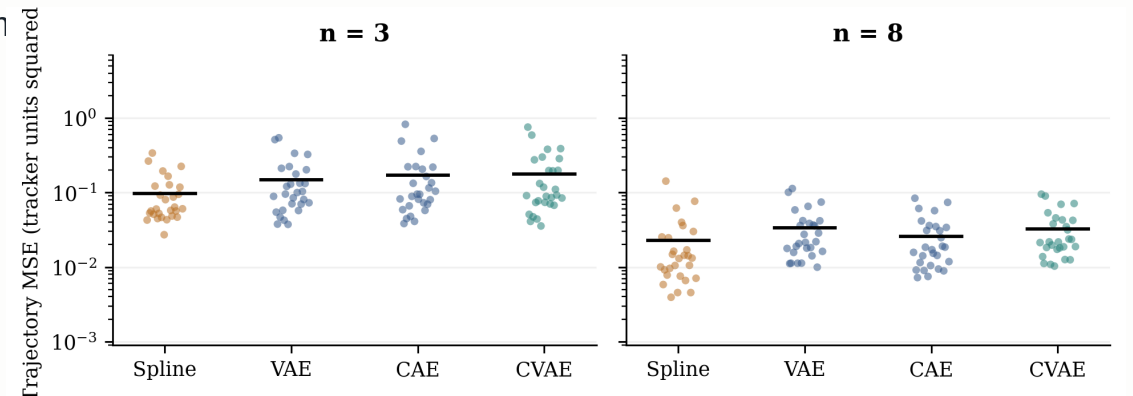
TASK 1 | RECONSTRUCTION

Can it redraw a movement it just saw?

Model	n=3	n=8
Spline + PCA	0.097	0.023
VAE	0.148	0.033
Condition-only Ridge	1.234	1.234

Participant-balanced MSE, tracker units squared. Lower is better.

- The linear baseline wins - and that is expected.
- PCA is optimal for squared error; the VAE trades accuracy for a san



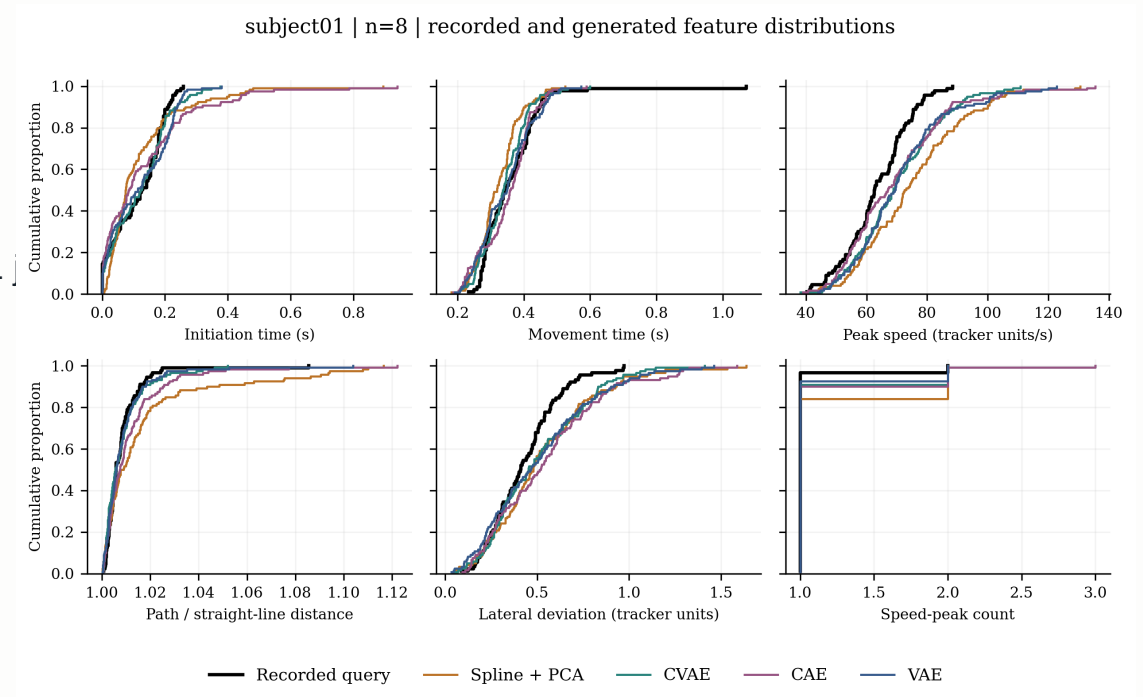
HOW WE MEASURE IT

What does 'matching a person' actually mean?

- Not trajectory-by-trajectory. We compare distributions.
- Every trial becomes 11 kinematic features:

initiation time . movement time . peak speed . time to peak
path length . straight-line distance . curvature . lateral deviation
speed-peak count . endpoint x . endpoint y

- Then: KS per feature, plus energy distance and MMD across all 11



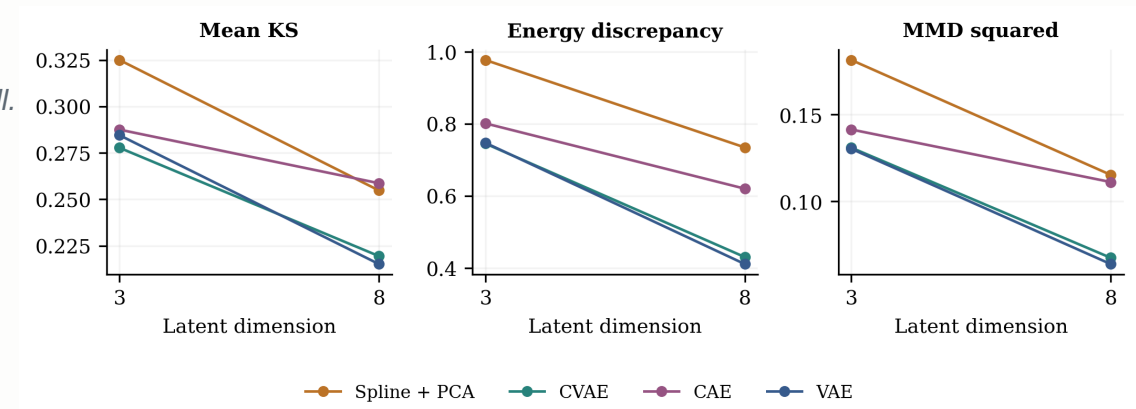
TASK 2 | GENERATION

Can it invent new movements for a person?

- Give the model ~85 of a person's trials. Never show it the rest.
- It generates 120 new trajectories. Compare against the held-out half.

Model (n=8)	Mean KS	Energy	MMD2
VAE	0.215	0.411	0.064
Spline + PCA	0.255	0.735	0.115

The VAE generates closer distributions - and it never sees the task conditions at all.



THE CONTROL THAT MATTERS

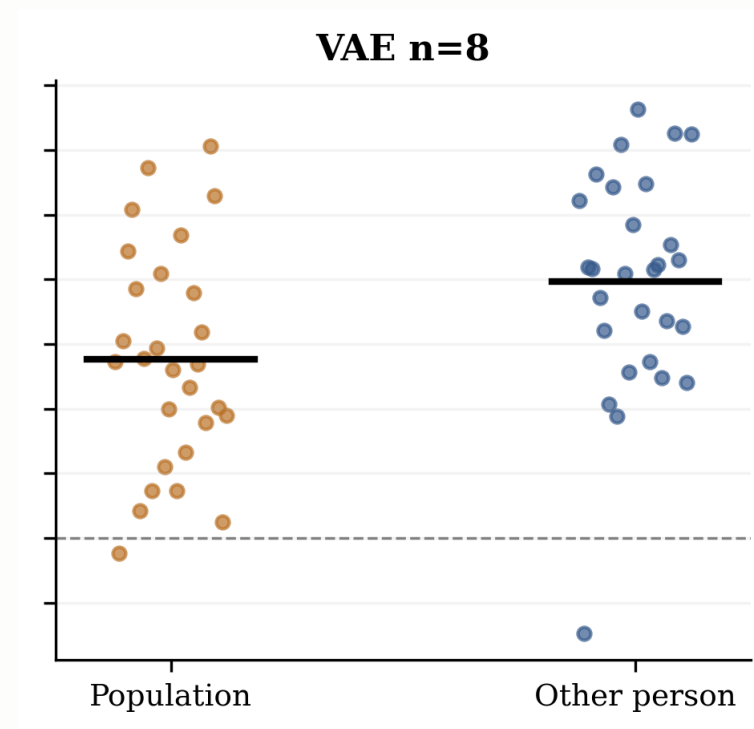
Does the RIGHT person's fingerprint matter?

- Freeze everything: decoder, task draws, noise, covariance, distance reference.
- Change one thing only - whose fingerprint we supply.

VAE n=8, fingerprint supplied	Mean KS	Energy	MMD2
Own participant	0.215	0.411	0.064
Population average	0.284	0.879	0.142
Another participant	0.314	1.168	0.195

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participants improve with their own fingerprint; all 6 contrasts $p < 1.5e-06$



HONEST SUMMARY

What holds, and what does not

Holds

- Personal information really is in the fingerprint - under a strict control.
- The VAE generates closer feature distributions than the baseline.

Does not

- Spline + PCA reconstructs better.
- The best generator ignores the task entirely.
- A simple average beats it on all 14 targets.

If you are building this today:

reconstruct with Spline + PCA; generate with the VAE. Use $n=8$ - $n=3$ is worse on almost every endpoint.

Caveat: n was matched for comparability, not chosen by a variance or elbow criterion. That criterion is the next step.

DEMO

See it work

- Pick a model and capacity, pick a participant.
- Move the latent sliders - watch the trajectory change.
- Compare generated against that person's real held-out trials.

